

DATA STRUCTURE

LAB PROGRAM

16. Write a C program to arrange a series of numbers using Insertion Sort

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int arr[100], n, i, j, key;
```

```
    // Read number of elements
```

```
    printf("Enter number of elements: ");
```

```
    scanf("%d", &n);
```

```
    // Read array elements
```

```
    printf("Enter %d numbers:\n", n);
```

```
    for(i = 0; i < n; i++)
```

```
    {
```

```
        scanf("%d", &arr[i]);
```

```
    }
```

```
    // Insertion Sort
```

```
    for(i = 1; i < n; i++)
```

```
{  
    key = arr[i];  
    j = i - 1;  
  
    while(j >= 0 && arr[j] > key)  
    {  
        arr[j + 1] = arr[j];  
        j--;  
    }  
  
    arr[j + 1] = key;  
}  
  
// Display sorted array  
printf("Sorted array:\n");  
for(i = 0; i < n; i++)  
{  
    printf("%d ", arr[i]);  
}  
  
return 0;  
}
```

Output

Enter number of elements: 5

Enter 5 numbers:

55

555

55555

5555

5

Sorted array:

5 55 555 5555 55555